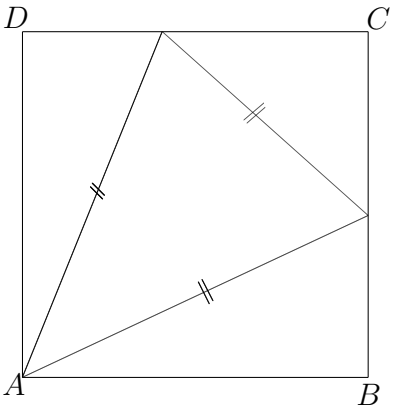
NMC 2011 O-LEVEL PAPER 1 QUESTIONS

SECTION A

Qn.1. The real numbers x, y and z satisfy $2^{x+y} = 10$, $2^{y+z} = 20$ and $2^{x+z} = 30$. Find the value of 2^x.

Qn.2. An equilateral triangle with sides of $10cm$ is inscribed in a square $ABCD$ in such a way that one vertex is at A, another vertex on $\overline{BC}$ and one on $\overline{CD}$. Find the area of the square.



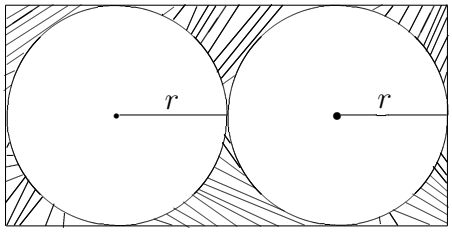
Qn.3. A prime-prime is a prime number that yields a prime when its units digit is omitted.(For example, 317 is a three-digit prime-prime because 317 is prime and 31 is prime). How many two-digit prime-primes are there?

Qn.4. Find the non-zero value of k such that the equation $x^2 + kx + k = 0$ has only one real solution.

Qn.5. Let Θ be defined by $x\Theta y = x^2 - y$ and $\oplus$ be defined by $x \oplus y = x + y^2$. What is the value of $(x \oplus y)\Theta y$?

Qn.6. Joshua was three times as old as Janet in 1993. In 2001, Joshua was 8 years older than Janet. How old is Janet now?

Qn.7. Find the area of the shaded region if each circle has radius r.



Qn.8. Given that $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots = y$. Find the value of $\frac{1}{10} + \frac{1}{20} + \frac{1}{40} + \frac{1}{80} + \ldots$ in terms of y.

Qn.9. The last pan cake is gone, and either Chris, Pat, Lou, Sam or Sam's dog ate it. The following is known. Each one (except Sam's dog) said they knew who ate the pan cake, Chris always tells the truth, Pat never tells the truth, Sam said it was either Pat, Chris or Lou. Lou said it wasn't me, Pat claimed Sam was a liar and Chris responded that Pat was the only liar. Who ate the last pan cake?

Qn.10. A shopper intends to buy 3 apples and 2 oranges and and will pay $5,000/=$. Changing his mind, he exchanges an orange for another apple and has to pay an additional $1,000/=$. What is the price of a single apple?

SECTION B : 10 **marks each**

Qn.11. Two numbers are written in base a as 32 and 24. The same two numbers are written in base b as 43 and 33, respectively. What is the sum of these two numbers in base 5?

Qn.12. Let a and b be different real numbers such that $a^2 = 6b + 5ab$ and $b^2 = 6a + 5ab$, find the value of ab.

Qn.13. If the 14 blocks below must be filled with whole numbers so that the sum of three consecutive blocks has a total of 18, find x.

			7				x				8		

Qn.14. A driver wishes to arrive at her destination at exactly $11:00$ $a.m.$ If she drives at 30 kmh, she would get there at $10:00$ $a.m.$ If she drives at 20 kmh, she would arrive at noon. How fast should she drive to arrive at $11:00$ $a.m$?

Qn.15. Find the sum of all positive values of x for which

$$\left(\log_{27} x^3\right)^2 = \log_{27} x^6.$$

NMC 2010 O-LEVEL PAPER 1 SOLUTIONS
SECTION A

1.

$$\begin{aligned}\log\left(\frac{(125)(625)}{25}\right) &= \log(125)(25)\\ &= \log_5\left(5^3\right)\left(5^2\right)\\ &= \log_5 5^5\\ &= 5\log_5 5 = 5.\end{aligned}$$

2.

$$\begin{aligned}\frac{a}{b}+\frac{b}{a}-ab &= \frac{a^2+b^2}{ab}-ab\\ &= \frac{a^2+b^2-(ab)^2}{ab}\\ &= \frac{a^2+b^2-(a-b)^2}{ab}\\ &= \frac{a^2+b^2-(a^2-2ab+b^2)}{ab}\\ &= \frac{2ab}{ab}=2.\end{aligned}$$

3. Let B and G denote the original number of boys and original number of girls respectively. Then

$$\frac{B}{G-15}=\frac{2}{1} \text{ and } \frac{B-45}{G-15}=\frac{1}{5}.$$

The two equations reduce to

$$B = 2(G-15) \quad (11)$$
$$5B = G+210. \quad (12)$$

Subtract equation (12) from $5\times$ (11) and get $10G-150-G-210=0$ from which we get $G=40$. Therefore, there were 40 girls at the beginning.

4.

$$\begin{aligned}
\text{Cost of one easy payment} &= \left(\frac{1}{4} \times 200,000\right) + 5,000 = UGShs\ 55,000.\\
\\
\text{Cost of three easy payments} &= UGShs\ 3 \times 55,000 = UGShs\ 165,000\\
\text{Total Cost of advertised product} &= \text{cost of three easy payments}\\
&+ \text{handling charges}\\
&= UGShs\ 165,000 + UGShs\ 15,000\\
&= UGShs\ 180,000.\\
\text{Money saved by buying from the television advertiser} &= UGShs\ (200,000 - 180,000)\\
&= UGShs\ 20,000.\\
\\
\text{Percentage saved} &= \frac{20,000}{200,000} = 10\%.
\end{aligned}$$

5. Let x and $2x$ be the ages in years of walter and his grandfather respectively at the end of 1994.

	Walter	Grandfather	Total
Age at the end of 1994	x	2x	3x
Year of birth	1994-x	1994-2x	3838

Since the sum of the years in which they were born is equal to 3838, we solve the equation $1994 - x + 1994 - 2x = 3838$ or $3988 - 3x = 3838$ from which we get $x = 50$ years. Therefore, at the end of 2010, Walter will be $50 + (2010 - 1994) = 66$ years.

6. Let m be the number of mangoes in Jesse's fridge originally.

	Mangoes at the beginning	Mangoes at the end
Day I	m	$m - 0.2m = 0.8m$
Day 2	$0.8m$	$0.8m - 0.2 \times 0.8m = 0.64m$

Therefore, $0.64m = 32$ implying that $m = 50$.

7. Let A represent numbers between 1 and 100 which are 1 less than a multiple of 5. Such numbers have unit digits of either 4 or 9.So,

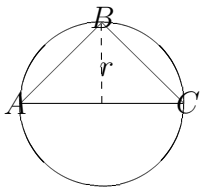
$$\begin{aligned}
A &= \{5n - 1 : n = 1, 2, \ldots, 20\}\\
&= \{4, 9, 14, 19, 24, 29, 34, 39, 44, 49, 54, 59, 64, 69, 74, 79, 84, 89, 94, 99\}.
\end{aligned}$$

Let B represent numbers between 1 and 100 which are 1 greater than a multiple of 4. Then,

$$\begin{aligned} B &= \{4m+1 : m = 1, 2, \ldots, 24.\} \\ &= \{5, 9, 13, , \ldots, 99\} \\ A \cap B &= \{\text{Those numbers in } A \text{ which are 1 greater than the multiple of 4}\} \\ &= \{9, 29, 49, 69, 89\}. \end{aligned}$$

Let C be prime numbers between 1 and 100. Then, $A \cap B \cap C$ are the prime numbers we can get from the set $A \cap B$. Therefore, $A \cap B \cap C = \{29, 89\}$ and their sum is $29 + 89 = 118$.

8. Of all triangles that are inscribable in the semi-circle ABC with the diameter AC as the base, the one with the greatest area is the one with largest height which is the radius of the circle. Therefore, area of such a triangle is $A = \frac{1}{2}bh = \frac{1}{2} \times 2r \times 2 = r^2$.



9. Let R be the annual rent. Expenditure: repairs and upkeep equal to $12.5\%R$ and the Amount Realised is $R - 12.5\%R - 650,000$. Since he earns 5.5% on his investment,it implies

$$\begin{aligned} 5.5\% \times 20,000,000 &= R - 12.5\%R - 650,000 \\ 1,100,000 &= 0.875R - 650,000 \\ 0.875R &= 1,750,000 \Rightarrow R = 2,000,000. \end{aligned}$$

Therefore, monthly rent is $\frac{2{,}0000{,}000}{12} = UGShs\ 166,666.67$.

10.

$$\begin{aligned} \text{Area of lune} &= \text{Area of semicircle } APB - \text{Area of segment } AB \\ &= \text{Area of semicircle } APB - (\text{Area of semicircle } APB - \text{Area of triangle } AOB\ ) \\ &= \text{Area of triangle } AOB = \frac{1}{2}r^2. \end{aligned}$$

The ratio of the area of the lune to the area of the triangle is 1 : 1.

11. (a) It takes Okello 9 days to do the work. Assuming that Okello works at a uniform rate, it implies that he does $\frac{1}{9}$ of the work in one day. Oryem is 50% more efficient than Okello. This means that in one day Oryem does 150% of the work done by Okello, which is $150\% \times \frac{1}{9} = \frac{1}{6}$. This it turn implies that Oryem takes 6 days to do the same piece of work if he works at a uniform rate.

    (b) Let l be the original length and w be the original width. The original area $A = lw$. Suppose that when w is increased by 10%, the length is decreased by d% to maintain the original area. Then

$$\begin{aligned} \text{New Area} &= \text{Original Area} \\ (100-d)l\% \times 110\%w &= lw \\ (100-d)\% \times 1.1lw &= lw \\ (100-d)\% \times 1.1 &= 1 \end{aligned}$$

    which implies that $d = 9.1$.

12. Let the speed of M be r. This implies that the speed of D is $r-4$. We have the following table.

	D	M
Distance in Miles traveled	48	72
Speed in miles per hour	$r-4$	r.

Since D and M travel for the same time, we have,

$$\begin{aligned} \text{Time taken by } D &= \text{Time taken by } M \\ \text{OR } \frac{48}{r-4} &= \frac{72}{r} \\ \text{OR } 48r &= 72(r-4) \end{aligned}$$

and this gives $r = 12$ miles per hour.

13. The even digits $2, 4, 6, 8$ all appear 20 times in between 1 and 100.
Their sum is $20(2+4+6+8) = 400$.

14. The four sums are

$$\begin{aligned} 7+a+b+1 &= k \quad (13) \\ 7+c+3 &= k \quad (14) \\ 3+e+f+10 &= k \quad (15) \\ 1+d+10 &= k \quad (16) \end{aligned}$$

Adding the four equations gives

$$42 + a + b + c + d + e + f = 4k. \tag{17}$$

since digits $2, 4, 5, 6, 8$ and 10 can be distributed among the letters a, b, c, d, e and f, we deduce from equation (17) that

$$42 + 2 + 4 + 5 + 6 + 8 + 10 = 4k$$

which implies that $k = 19$.

From equation (14), we have that $7 + c + 3 = k$ which implies that $c = k - 10 = 19 - 10 = 9$.

From equation (16), we have that $1 + d + 10 = k$ which implies that $d = k - 11 = 19 - 11 = 8$.

From equation (13), we have that $7 + a + b + 1 = k$ which implies that $a + b = k - 8 = 11$.

Looking at the remaining digits, we can let $a = 5$ and $b = 6$.

From equation (15), we have that $3 + e + f + 10 = k$ which implies that $e + f = k - 13 = 6$. Looking at the remaining digits, we can let $e = 2$ and $f = 4$.

15. Let $\overline{DC} = x$. Since $\triangle CDE$ is a right angled triangle with $\angle DCE = 60^0$, we have $\overline{CE} = \frac{\overline{EC}}{\cos 60} =$

$2\overline{EC} = 2x$ and $\overline{DE} = \cos 30 \times 2x = \sqrt{3}x$. So, $\overline{FE} = \overline{DE} = \sqrt{3}x$. Since $\angle FEA = 90^0$, we have $\overline{AE} = \frac{\overline{FE}}{\tan 60} = \frac{\sqrt{3}x}{\sqrt{3}} = x$. So, $\overline{AE} = \overline{DC}$ and $\overline{AC} = \overline{AE} + \overline{EC} = 3x$. Therefore,

$$\begin{aligned} \text{Area of } \triangle ABC &= \frac{1}{2}\overline{BA} \times \overline{BC} \times \sin 60 \\ &= \frac{1}{2} \times 3x \times 3x \times \frac{\sqrt{3}}{2} = \frac{9\sqrt{3}}{4}x^2 \end{aligned}$$

and

$$\begin{aligned} \text{Area of } \triangle DEF &= \frac{1}{2}\overline{FD} \times \overline{FE} \times \sin 60 \\ &= \frac{1}{2} \times \sqrt{3}x \times \sqrt{3}x \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}x^2. \end{aligned}$$

$$\begin{aligned} \text{Ratio of Area of } \triangle DEF : \text{Area of } \triangle ABC &= \frac{3\sqrt{3}}{4}x^2 : \frac{9\sqrt{3}}{4}x^2 \\ &= 1 : 3. \end{aligned}$$